

The Simple Economics of Recursive Self-Improvement

Tom Cunningham^{*}

Lukas Althoff[†] Basil Halperin[†] Brian Jabarian[†] Andrew Koh[†]
Arjun Ramani[†] Phil Trammell[†] Parker Whitfill[†] Cheryl Wu[†]

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Abstract

We model the economics of recursive self-improvement (RSI) and assess its plausibility and impacts. First, we develop a sequence of increasingly rich models of AI progress to highlight the key feedback loops behind RSI. We extend existing theoretical work to distinguish between “narrow” and “broad” AI capabilities, capturing the possibility that capabilities first accelerate narrowly in verifiable domains before expanding to less verifiable ones. Second, we calibrate our model to existing estimates of the main parameters behind our models. In doing so, we put together a wish list of empirical objects that AI companies and analysts should endeavor to measure and release to the public. Based on these data, we discuss the plausibility of a *self-sustaining acceleration* in AI capabilities and its possible economic impacts.

Contributions: ^{*} Lead author. [†] Listed in alphabetical order; these authors contributed equally.

Corresponding author: Tom Cunningham, tom.cunningham@metr.org

Latest version of the paper: <https://github.com/elasticity-ai/elasticity/raw/main/paper/elasticity-rsi-paper.pdf>

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1 INTRODUCTION

Motivation. The continuous improvement of technology is the cornerstone of modern economic growth. Each industrial revolution has been fueled not by a single technological breakthrough but by a virtuous cycle of thinkers and tinkerers iteratively improving technologies (Mokyr, 2002). For the first time, a general-purpose technology may learn to do the thinking and tinkering itself: artificial intelligence (AI).

Core argument. AI researchers at frontier labs report that AI systems are increasingly involved in AI research (Favaro and Clark, 2026). This has caused some researchers to expect an *acceleration* in the growth rate of AI capabilities, relative to the trend (measured, for example, via the Epoch Capability Index or METR time horizon – though measuring capabilities comprehensively remains difficult).

The degree of the acceleration in AI capabilities depends on what we call the “core feedback loop”: for a one unit increase in model capabilities, how much do the capabilities of the next generation of models increase? Estimating the strength of this relationship is challenging because it is governed by many inputs and bottlenecks.

This note presents a series of theoretical models to clarify the forces behind the core feedback loop. We draw a tight connection to empirical data needed to measure the strength of the feedback loop, which, under our model’s assumptions, can help assess the degree of current and future acceleration.

Defining RSI. The term “recursive self-improvement” (RSI) has been used in several ways, often inconsistently. Some define it as a technology contributing to its own improvement, which could apply to almost any technology since the dawn of human civilization.¹ Anthropic recently defined RSI more narrowly as “an AI system capable of fully autonomously designing and developing its own successor”. Other definitions highlight the importance of acceleration or feedback loops, even if full automation is not realized.

We remain agnostic to what constitutes RSI and instead focus on *self-sustaining acceleration* in AI capabilities and derive the conditions under which it arises. We begin by defining self-sustaining acceleration and related terms in Table 1.

Table 1: Definitions and closest references for AI feedback and takeoff concepts

Term	Definition
Feedback loops	When higher capability levels increase future capability levels, holding other inputs fixed (human R&D, capital, etc.). Arguably most technologies exhibit feedback effects, and AI has certainly exhibited a degree of feedback for a long time.

¹The origin of the term is not clear, but it was popularized by (and plausibly originated with) Yudkowsky (2001, 2007, 2008), extending an argument by Good (1965) .

Table 1: Definitions and closest references (continued)

Term	Definition
R&D automatability	When AI is able to autonomously make technological improvements at least equivalent to those made by unaided human researchers, at an equal cost. Even when this is true, humans could still be employed in R&D due to slow diffusion, regulation, or complementarities.
Self-sustaining acceleration	When AI systems sustain accelerating progress in AI capabilities without any growth in exogenous inputs (human labor, training compute, etc.).
Intelligence explosion	When AI capabilities go to infinity in finite time.

References: Davidson et al. (2026), Chan et al. (2026), Eth and Davidson (2025b), Davidson and Houlden (2025), Davidson (2023), Shulman and Sandberg (2010), Good (1965), Hanson (2008), Christiano (2018), Ho and Whitfill (2025), Aghion et al. (2017).

Those concepts are often conflated with each other and the term “RSI.” Most commonly, we see AI commentators assume that full automation of AI R&D *necessarily* leads to self-sustaining acceleration. This conflation follows a rich tradition. For example, I.J. Good in 1965 states: “an ultraintelligent machine could design even better machines; there would then *unquestionably* be an intelligence explosion”. But our models make clear that this does not necessarily follow if ideas get harder to find, or if feedback loops get bottlenecked.

Modeling RSI. We construct a sequence of progressively richer models of RSI. We present the models in diagrams rather than emphasizing equations, for readability.

Our basic model distinguishes three production functions that combine to form a feedback loop: the production of AI capabilities, algorithmic efficiency, and R&D effort. A fourth production function, for output, determines economic impacts. We then extend the model to allow for bottlenecks, where the necessity of humans may break the feedback loop; for economic feedback loops, where higher output finances further compute; for experimental compute; and for data. Finally, we distinguish narrow from broad capabilities, allowing RSI affect easier-to-verify tasks (coding, math, benchmark optimization) without spilling over into harder-to-verify tasks.

Our key technical contribution is to introduce a simple graphical framework to represent an otherwise complicated system of variables and feedback loops. Nodes of the graph represent outputs of a production function; the strength of edges along the graph represent elasticities of an output with respect to inputs; and the self-sustaining acceleration condition is derived from multiplying across the strengths of these edges for each feedback loop and summing over loops. Details of this framework are outlined in technical boxes.

Existing empirical evidence & requests for data. Our models depend on a set of critical measurable parameters. We review existing empirical measures of these parameters and put forward a list of measures that would be useful for AI companies to provide. The biggest unknowns to be informed by additional data are (i) the growth rate of algorithmic efficiency in labs and (ii) lab expenditures across inputs (human R&D, data, experimental compute, inference for R&D), and (iii) direct measures of how much models contribute to research like the share of technical advances produced by AI.

Plausibility and implications. The strongest evidence in favor of a self-sustaining acceleration is the combination of high estimated returns to research, relative to R&D in the whole economy (Bloom et al., 2020), and rapidly advancing AI capabilities on coding and research benchmarks. The strongest evidence against is that capability growth is narrow but broad capabilities are necessary for continued R&D effort, algorithmic progress has historically been tied to compute scaling, and continued compute scaling may itself become constrained by GDP or power constraints. Political and social barriers are likely to slow deployment even if the technical conditions for self-sustaining capability acceleration are met.

Position in the literature. The economics literature on RSI is founded on Aghion et al. (2017), who derived conditions under which AI-driven automation could generate accelerating or even explosive economic growth, and Davidson (2023), whose compute-centric framework offered an early formal model of AI takeoff dynamics. We complement an emerging economics literature on RSI in three ways. First, Davidson et al. (2026) formally develops a semi-endogenous growth model with an innovation network, and Eth and Davidson (2025b) calibrates a model of a software-only takeoff. Our framework is deliberately simple and often informal, allowing us to be more wide-ranging. Second, following Cunningham and Shetty (2026), we emphasize the distinction between narrow and broad capabilities in both theory and the discussion of economic impact. This distinction is largely set aside in the review articles of Trammell and Korinek (2025) and Jones (2026). Third, we provide a wish list of empirical objects that AI companies can both measure and feasibly share openly that would be informative for the public conversation on RSI. This supplements the existing empirical efforts that we review (e.g. Whitfill and Wu 2025, Gundlach et al. 2025, Epoch 2026).

2 A SIMPLE MODEL OF RECURSIVE SELF-IMPROVEMENT

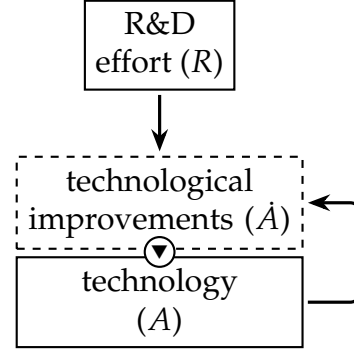
In this section we present a sequence of models to understand the economics of recursive self-improvement. We represent the models with diagrammatic graphs, rather than emphasizing equations, to keep the focus on the general feedback loops rather than any specific parametric structure. Nonetheless, we are careful to ensure that these diagrams can be understood formally as *directed graphs* that map precisely to well-defined economic models. In a technical details box, for interested readers only, we make precise how to translate our graphs into equations.

2.1 Jones Model of Innovation

In the standard model of innovation of Jones (1995), two channels affect technological improvements ($\dot{A}$):

1. R&D effort (R) such as labor and capital contribute to new technological improvements ($\dot{A}$).
2. Technological improvements also build on the existing stock of technology (A), subject to potentially diminishing returns (“ideas getting harder to find”) or increasing returns (“new technology makes finding new ideas easier”).

Figure 1: Jones Model of Innovation



Importantly, there is a *feedback loop* between the stock of technology and new technological improvements: technological improvements increase the level of technology (by definition), and a higher level of technology affects further technological improvements.

The strength of this feedback loop, in the Jones model, is characterized by the responsiveness—that is, the *elasticity*—of technological improvements to the stock of technology. In the figure, this is the strength of the arrow from A to $\dot{A}$, and it is denoted by $\varepsilon_{\dot{A}, A}$.

There is self-sustaining acceleration in technology in the Jones model if this feedback loop is sufficiently strong: in particular, if $\varepsilon_{\dot{A}, A} > 1$. Empirically, in traditional industries ranging from agriculture to semiconductors, this condition does not hold: technological improvements are getting harder to find (Bloom et al., 2020).

Elasticity

An elasticity measures responsiveness in proportional terms: the percentage change in an output induced by a one-percent change in an input, holding other inputs fixed. It is unit-free, so it does not depend on how either variable is scaled or measured.

Formal definition. For a variable $Y = f(X_1, \dots, X_n)$, the elasticity of Y with respect to X_j is

$$\varepsilon_{Y, X_j} \equiv \frac{\partial \log Y}{\partial \log X_j} = \frac{\partial f}{\partial X_j} \frac{X_j}{Y}.$$

Equivalently, it is the local slope of Y against X_j on logarithmic axes. For a power function $Y = X^a$, the elasticity is constant and equal to the exponent a .

Technical details: graphs

What do graph edges mean? For each model in this section, we use a directed graph to precisely represent how economic variables feed into each other. Each variable is entirely determined by its parents. For instance, in the canonical Jones model above both technology (A) and R&D effort (R) determine the rate of technological improvements. That is, $\dot{A} = f(A, R)$ for some function f and we write

$$\varepsilon_{\dot{A},A} = \frac{\partial \log f(A, L)}{\partial \log A} \quad \text{and} \quad \varepsilon_{\dot{A},L} = \frac{\partial \log f(A, L)}{\partial \log L}$$

where $\varepsilon_{\dot{A},L}$ is the returns to scale in contemporaneous research effort and $\varepsilon_{\dot{A},A}$ is the rate of diminishing returns to R&D.

Our directed graphs distinguish stock variables from flow variables to ensure that the diagrams map one-to-one to our system of equations without ambiguity (Richardson, 1986). We use solid arrows ($\longrightarrow$) to represent how variables relate to each other at each point in time. For instance in the model above we have $A \longrightarrow \dot{A}$ and $R \longrightarrow \dot{A}$ so $\dot{A}$ at every point in time is pinned down by its parents A and R . The rate of change of a stock variable (e.g. $\dot{A}$) accumulates into itself (e.g. into A) over time, which we denote with triangles ($\blacktriangleright$).

Self-sustaining acceleration. The growth path of technology A is governed by the local elasticity of its time derivative $\dot{A}$ with respect to A :

$$\varepsilon_{\dot{A},A} \equiv \frac{d \log \dot{A}}{d \log A} = \left(\sum_{\substack{\text{paths } \mathbf{e} \\ \text{from } A \text{ to } \dot{A}}} \prod_i \varepsilon_{e_i} \right).$$

where paths are ordered sequences of directed edges $\mathbf{e} = (e_1, \dots, e_i, \dots)$. This equation highlights that the contribution of each path to the growth in efficiency is determined by product of elasticities along the path's edges by the chain rule. This elasticity determines how the growth rate of capabilities $g_A = \dot{A}/A$ changes as A grows, yielding three regimes:

$$\begin{cases} \varepsilon_{\dot{A},A} < 1 : & g_A \text{ falls as } A \text{ grows} \Rightarrow \text{sub-exponential growth (fizzling out).} \\ \varepsilon_{\dot{A},A} = 1 : & g_A \text{ constant} \Rightarrow \text{exponential growth (knife-edge).} \\ \varepsilon_{\dot{A},A} > 1 : & g_A \text{ rises as } A \text{ grows} \Rightarrow \text{super-exponential growth (accelerates).} \end{cases}$$

We call growth *self-sustaining acceleration* when $\varepsilon_{\dot{A},A} > 1$, i.e. when g_A rises over time with no change in exogenous inputs.

In Figure 1 above, we have a single feedback path $A \rightarrow \dot{A}$, so growth would show self-sustaining acceleration if $\varepsilon_{\dot{A},A} > 1$.

Self-sustaining acceleration at one particular point in time (locally self-sustaining acceleration) does not necessarily imply self-sustaining acceleration forever (i.e., globally self-sustaining acceleration) because the elasticities are not necessarily constant. We strongly suspect this is the case for the elasticities we consider: for example, the elasticity of research effort to AI capabilities ($\varepsilon_{R,C}$) discussed in our baseline

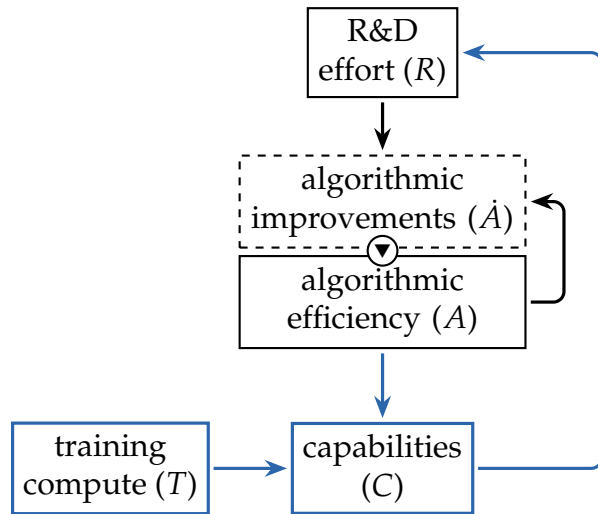
model below was plausibly near zero at the GPT-1 level but is substantially higher now. But even local self-sustaining acceleration could lead to a substantial change in the pace of AI progress.

2.2 Baseline Model of RSI

To understand progress in AI capabilities (C), we turn from studying “technology” to studying *algorithmic efficiency*: the efficiency of AI models given a fixed stock of training compute. Our baseline model distinguishes between four different production functions.

1. Algorithmic improvements ($\dot{A}$) increase with R&D effort (R), and depend on the existing level of algorithmic efficiency (A).
2. Algorithmic efficiency (A) grows with algorithmic improvements ($\dot{A}$).
3. AI capabilities (C) are determined by algorithmic efficiency (A) and the amount of training compute (T).
4. R&D effort (R) is increased by AI capabilities, (C).

Figure 2: Baseline Model of RSI



Critically, the fact that AI capabilities now feed into R&D effort creates a new, second feedback loop. Self-sustaining acceleration in algorithmic efficiency now depends on the strength of *two* feedback loops:

1. The loop between algorithmic improvements and algorithmic efficiency, analogous to the loop in the Jones model.
2. The loop where stronger AI capabilities increases R&D effort, allowing for new algorithmic improvements, thus improving algorithmic efficiency, which strengthens AI capabilities yet further. This is the *core feedback loop* throughout our models of recursive self-improvement.

Algorithmic efficiency exhibits self-sustaining acceleration if the sum of the strengths of the feedback loops, each given by the sum of the strengths of the arrows in that loop, exceeds 1. This is empirically measurable, as discussed in section 3.

Technical details: baseline model

Self-sustaining acceleration. The capability feedback loop adds a path from A to $\dot{A}$ through C on top of the direct spillover from A to $\dot{A}$. Thus the total elasticity of algorithmic improvements with respect to A is

$$\frac{d \log \dot{A}}{d \log A} = \underbrace{\varepsilon_{\dot{A}, A}}_{\text{direct spillover } A \rightarrow \dot{A}} + \underbrace{\varepsilon_{\dot{A}, R} \varepsilon_{R, C} \varepsilon_{C, A}}_{\text{capability loop } A \rightarrow C \rightarrow R \rightarrow \dot{A}} \geq 1$$

By the criterion above, growth shows self-sustaining acceleration (g_A increases as A grows) when this exceeds one.

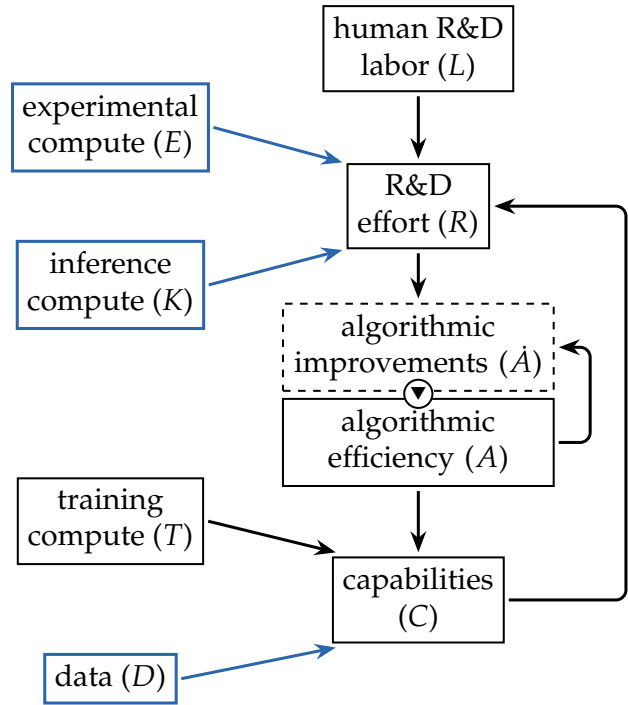
2.3 Compute & Data Bottlenecks

We now extend the production functions for R&D effort (R) and AI capabilities (C) as follows:

1. R&D effort (R) now has three inputs: human R&D labor (L), experimental compute (E), and inference compute (K).
2. AI capabilities (C) now has three inputs: algorithmic efficiency (A), training compute (T), and data (D).

There are still two feedback loops ($A \rightarrow \dot{A}$ and $A \rightarrow C \rightarrow R \rightarrow \dot{A}$). However, the inputs into the production functions can be complements or substitutes which may impact growth. For R&D effort, if experimental compute is a complement to AI labor in discovery, then $\varepsilon_{R, C}$ may fall over time as research effort becomes increasingly bottlenecked by experimental compute (Whitfill and Wu, 2025; Ho and Whitfill, 2025).

Figure 3: Model with Bottlenecks



If training compute is a complement to algorithmic efficiency, then $\varepsilon_{C, A}$ might fall over time, as capabilities become bottlenecked by training compute (Gundlach et al., 2025). Data is also a critical input to the AI capabilities production function. Unlike narrow

domains that have seen spectacular progress via synthetic data produced by inference compute, lack of real-world data may bottleneck broad RSI (Millidge, 2025) and require learning-by-doing via real-world deployment (Farboodi et al., 2025; Erdil et al., 2025). If these inputs are bottlenecks, then the self-sustaining acceleration of A is no longer sufficient for self-sustaining acceleration in C , though it remains necessary.

Technical details: bottlenecks

Self-sustaining acceleration. The self-sustaining growth condition for A is the same as the prior model

$$\underbrace{\varepsilon_{A,A}}_{\text{direct spillover } A \rightarrow \dot{A}} + \underbrace{\varepsilon_{A,R} \varepsilon_{R,C} \varepsilon_{C,A}}_{\text{capability loop } A \rightarrow C \rightarrow R \rightarrow \dot{A}} \geq 1$$

However, if the other inputs to R&D effort are necessary (e.g. humans for physical labor or experimental compute for running experiments) then that implies the elasticity of R&D effort to AI capabilities, $\varepsilon_{\dot{R},C}$ will fall over time as further progress is bottle necked.

Aside on AI capabilities $\rightarrow$ research input Note that human researchers (L), AI capabilities (C) and experimental compute, E , combine to produce R&D effort (R). Previous literature, such as Eth and Davidson (2025a), has tried to parametrize this function. But our view is that it is not obvious what this function looks like because it depends on “the returns to intelligence” and how humans and AI systems differ, both of which we have a poor understanding of. Luckily the only thing we need to know is the elasticity noted above and it may be possible to measure this elasticity without fully understanding AI capabilities. We will discuss this further in the data section.

2.4 Economic Feedback Loops

The previous model implicitly held constant the level of compute available for training, inference, and experiments as well as holding constant. This isolated the role of algorithmic efficiency progress in driving a software-only singularity. There are also economic feedback loops: AI capabilities driving higher GDP, which in turn finances further investment in compute and R&D, raising AI capabilities (Davidson et al., 2026):

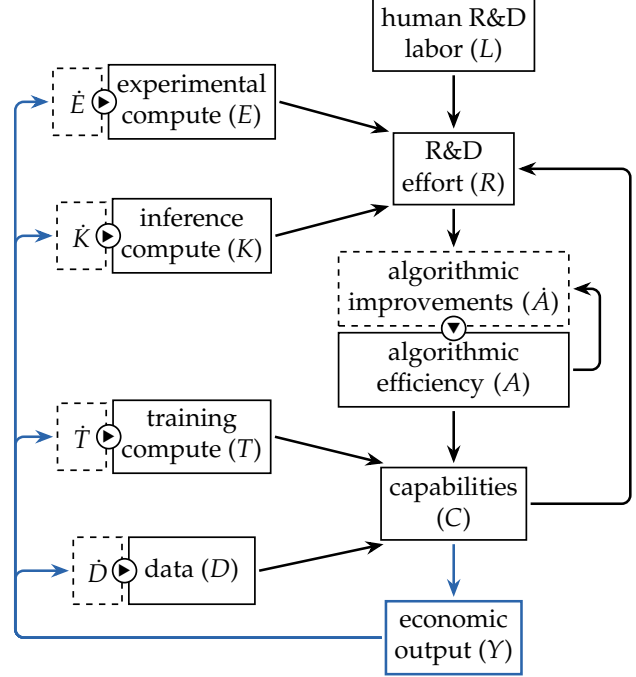
1. Increased AI capabilities (C) drive higher economic output (Y).
2. Higher GDP in turn finances further investment in compute (for experiments E , for inference K , and for training T), which feeds back into AI capabilities.

The previous model also held constant the quantity of data (D) but as AI capabilities drive economic production, they might generate more data increasing AI capabilities further (Farboodi et al., 2025).

Aside from the two feedback loops before ($A \rightarrow \dot{A}$ and $A \rightarrow C \rightarrow R \rightarrow \dot{A}$), self-sustaining acceleration in algorithmic efficiency now depends on the strength of more economic feedback loops that run through AI capabilities. For instance, capabilities might drive the accumulation of high-quality data which in turn drives capabilities ($C \rightarrow Y \rightarrow \dot{D} \rightarrow D \rightarrow C$) or the accumulation of training compute which in turn drives capabilities ($C \rightarrow Y \rightarrow \dot{T} \rightarrow T \rightarrow C$). This implies that self-sustaining acceleration in algorithmic efficiency is now neither necessary or sufficient for self-sustaining acceleration of capabilities. To derive the condition for self-sustaining acceleration, we require taking into account these more complicated feedback loops.

We emphasize that although economic feedback loops suggest a macroeconomic channel (e.g. Y as GDP), we might alternatively interpret Y as the profits of a single AI firm. As AI capabilities progress and the firm grows more profitable, they are able to afford more chips or data (without substantially altering factor prices). Such feedback loops are

Figure 4: Model with Economic Feedback Loops



already active, and we expect it to drive capability acceleration even without substantial macroeconomic impacts.

2.5 Narrow vs Broad Capabilities

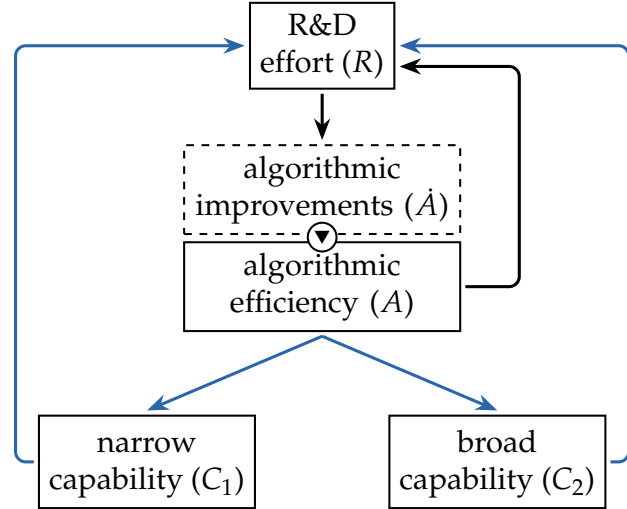
Suppose there are two types of capability, narrow (C_1) and broad (C_2), that each rise with efficiency A but with different elasticities (θ_1, θ_2). Each also contributes to algorithmic improvements $\dot{A}$ with different weights (χ_1, χ_2).

$$\dot{A} = L^\lambda A^{1-\beta} C_1^{\chi_1} C_2^{\chi_2},$$

$$C_1 = A^{\theta_1}, \quad C_2 = A^{\theta_2}.$$

This maps onto the “low-hanging fruit” view of AI R&D (Cunningham and Shetty, 2026): narrow capabilities (C_1) cover the easily-verified, quickly-saturated tasks that AI improves first, while broad capabilities (C_2) cover the harder, less-verifiable tasks. Whether growth shows self-sustaining acceleration then depends on whether the narrow channel ($\chi_1\theta_1$) keeps delivering before the broad channel ($\chi_2\theta_2$) is needed.

Figure 5: Model that Distinguishes Narrow versus Broad Capabilities



The narrow-broad distinction can be made in several ways: tasks with output which is “verifiable”, versus those in which it is not; tasks which are not “messy”, versus those which are; those not requiring “taste”, versus those which do. Although the preceding diagrams abstracted away from this distinction, we think it will be crucial. There is suggestive, growing evidence that advances in capabilities are occurring more quickly in narrow domains (Cunningham and Shetty, 2026). If AI R&D has differential impacts on the progress of narrow and broad capabilities, this might bottleneck improvements in algorithmic efficiency. For example, the lack of ‘research taste’ in AI systems might prevent compute from fully substituting for researcher labor, which could delay or even preclude rapid progress in broad capabilities. Yet, we also emphasize that narrow and broad lie on a spectrum. This means that what starts out as ‘merely’ a narrow capability acceleration might eventually drive broader improvements as capabilities advance step-by-step up the narrow→broad ladder (Farboodi et al., 2025): improvements in performance on ‘routine tasks’ could boost the pace of algorithmic improvements which, in turn, drive improvements in performance on ‘substantial tasks’ which, by further boosting algorithmic improvements, drive improvements in performance on ‘open-ended problems’.

	Narrow RSI	Broad RSI
Small econ impact	AI capabilities are astounding as measured by benchmarks containing cheap-to-verify tasks, but the wider economy is bottlenecked by messier hard-to-verify tasks	Unlikely, but could occur because of regulation
Large econ impact	AI capabilities as measured by benchmarks with cheap-to-verify tasks are representative of true economic capabilities	Economically transformative

3 DATA RELEVANT TO RSI

RSI is most consequential if it leads to rapid capabilities growth over short time horizons. This makes our bottlenecks model above the most relevant, where AI might be able to self-improve its algorithms, but data, compute, and other complementary inputs remain approximately fixed in the short run. Therefore, we focus this data section on the self-sustaining acceleration condition of the bottleneck's model:

$$\underbrace{\varepsilon_{\dot{A},A}}_{\substack{\text{direct} \\ \text{spillover} \\ A \rightarrow \dot{A}}} + \underbrace{\varepsilon_{\dot{A},R} \varepsilon_{R,C} \varepsilon_{C,A}}_{\substack{\text{capability loop} \\ A \rightarrow C \rightarrow R \rightarrow \dot{A}}} \geq 1,$$

We leave discussion of data calibration of the larger model with economic feedback loop to further research as it is both more complicated and less urgent.

Taking the self-sustaining acceleration condition from the bottlenecks model above and re-arranging gives:

$$\underbrace{\frac{\varepsilon_{\dot{A},R}}{1 - \varepsilon_{\dot{A},A}}}_{\text{return to R\&D}} \cdot \underbrace{\varepsilon_{R,C}}_{\substack{\text{research effort's} \\ \text{elasticity to} \\ \text{AI capabilities}}} \cdot \underbrace{\varepsilon_{C,A}}_{\substack{\text{capabilities'} \\ \text{elasticity to} \\ \text{algorithmic} \\ \text{progress}}} \geq 1.$$

In our view, estimating this product, tracking it over time, and extrapolating when it might cross 1 is the single most important empirical exercise for assessing whether a capabilities explosion will occur. The rest of this section maps each term in the condition to the data needed to estimate it: Section 3.1 covers the return to research, Section 3.2 covers the elasticity of research effort with respect to AI capabilities, and Section 3.3 the response of capabilities to algorithmic progress, the term most directly estimated by existing scaling-law estimates. We then collect supplementary detailed evidence on model-training inputs and on model economics to build a broader picture of AI R&D automation.

3.1 Return to Research

Although it is not immediately obvious from the diagram, the underlying model implies that the first term, the “returns to research”, can be measured by combining two terms: the growth rate of algorithmic efficiency ($g_A \equiv \dot{A}/A$) and the growth rate of R&D effort ($g_R = \dot{R}/R$):

$$\underbrace{\frac{\varepsilon_{\dot{A},R}}{1 - \varepsilon_{\dot{A},A}}}_{\text{return to R\&D}} = \frac{g_A}{g_R}$$

This holds exactly on a balanced growth path, i.e. when algorithmic progress g_A is constant, or approximately when it is approximately constant.² The smooth and steady historical rate of algorithmic progress makes this balanced-growth approximation a reasonable benchmark, though not one we expect to hold further into the future if RSI pushes us off the balanced-growth path. Therefore, we need to estimate g_A and g_R .

For g_A , we measure the rate of algorithmic progress directly from model-training data. For g_R , we can use the identity $g_R = \sum_{\text{input}} \varepsilon_{R,\text{input}} g_{\text{input}}$.³ Because elasticities equal cost shares under cost-minimizing input choice,⁴ we can re-express this as $g_R = \sum_{\text{input}} S_{\text{input}} g_{\text{input}}$ where S_i denotes the share of research spend on input i . We therefore need the cost share and growth rate of each input to R&D effort: human labor L , experimental compute E , and inference compute K .

Model primitive	Existing data	Data ask
Growth in algorithmic efficiency (g_A)	Ho et al. (2024) estimate pretraining algorithmic progress of 3x a year (i.e. compute needed to achieve a given level of capability is 1/3 as much each year) Whitfill et al. (2025) estimate 3.5x/year for the entire stack. Ho (2026) estimates 10x/year for the entire stack. Gundlach et al. (2025) argue that algorithmic progress has grown at different rates across scales: 1.5x/year at small scale, and 2.5x/year at frontier scale, for pretraining.	Current data weakness: Compute estimates for models since 2024 are uncertain. Ideal new data: model training compute after 2024 to estimate g_A.

²To see this, log-differentiate $g_A = \dot{A}/A = f(R, A)/A$, which gives $d \log g_A / dt = \varepsilon_{\dot{A},R} g_R - (1 - \varepsilon_{\dot{A},A}) g_A$. Constant g_A zeroes the left side, so $g_A/g_R = \varepsilon_{\dot{A},R}/(1 - \varepsilon_{\dot{A},A})$. Empirically, g_A has mostly been constant so far.

³This is the growth-rate version of the chain rule: $g_R = \sum_i \varepsilon_{R,i} g_i$ summing over R&D inputs.

⁴It is without loss of generality to assume R has constant returns to scale, since R is a latent index and the elasticity term controls its returns to scale. If firms optimize, elasticities equal cost shares. If firms make mistakes, the cost shares are at least their revealed-preference belief about the elasticities.

Model primitive	Existing data	Data ask
Share of human researchers (L) of R&D spend, and its growth g_L Its share S_L and growth g_L enter $g_R = \sum_{\text{input}} S_{\text{input}} g_{\text{input}}$.	Epoch AI (2026) estimates growth rates of around 2–3x/year in total staff for frontier labs over 2023–2025. Whitfill and Wu (2025) estimate headcount growth rates of 2–3X/year for Anthropic and OpenAI over 2022–2024. Cottier et al. (2024) estimates that around 1/3 of the cost of training models was spent on researcher salaries.	Current data weakness: Employee headcount does not necessarily map to the researcher headcount. Ideal new data: Granular lab data on staff, split by seniority and by researchers vs. engineers, would let us treat these as separate inputs rather than an aggregate L.
Share of experimental compute (E) of R&D spend, and its growth g_E Its share S_E and growth g_E enter $g_R = \sum_{\text{input}} S_{\text{input}} g_{\text{input}}$.	You (2025) estimates that in 2024 only 10% of OpenAI’s R&D compute was on final training runs, while 90% was on experimental compute (i.e. training which did not directly affect weights used in a public product). Denain and Wu (2026) estimate similar ratios for Chinese companies. Morrison et al. (2026) estimate similar ratios for OLMo 3 family. You et al. (2026) finds that the total growth rate of chips is about 3.4x a year.	Current data weakness: experimental compute share in total compute is not experimental compute share in total R&D expense. The composition of experimental compute is also not clear. Ideal new data: Labs could publish detailed data on experimental compute spend across domains and over time, which would help estimate how much experimental compute bottlenecks automated research.
Share of inference compute (K) of R&D spend, and its growth g_K Its share S_K and growth g_K enter $g_R = \sum_{\text{input}} S_{\text{input}} g_{\text{input}}$.	No data available.	Ideal new data: Labs could publish detailed data on inference compute spend across domains and over time.

Model primitive	Existing data	Data ask
Substitutability of labor and experimental compute How substitutable labor and compute are controls how fast $\varepsilon_{R,C}$ falls as research becomes bottlenecked by experimental compute.	Whitfill and Wu (2025) try to estimate the substitutability between labor and experimental compute in AI R&D, using variation in relative prices, but their findings are inconclusive.	Current data weakness: Aside from data quality, it is unclear whether the need for experimental compute increases with higher training compute. Ideal new data: labs could conduct experiments that would give teams randomized amounts of R&D labor and experimental compute, but this is expensive. One cheaper substitute is using AI labor in place of human labor in the experiment. Cheaper still is to survey managers how much more progress they would make with (A) twice the researchers (compute fixed); (B) twice the compute (researchers fixed); (C) both doubled.

3.2 Research Effort to Capabilities

The second term measures how strongly effective research effort responds to more capable AIs. This is the hardest of the three to measure, since it captures how AI capabilities translate into real-world research effort.

One possible route is to measure it directly, through studies of how much AI accelerates end-to-end research.

A second route uses a cost-share argument: suppose that AI enters R&D effort by a capabilities adjusted measure of the number of researchers. That is, let KC^γ denote the number of effective researchers where K denotes quantity of inference compute and C^γ denotes some quality adjustment - and $R(\cdot, \dots, KC^\gamma, \dots)$. Then $\varepsilon_{R,C} = \gamma S_k$ by a modified version of the elasticity-cost minimization correspondence.

Model primitive	Existing data	Data ask
Share of AI R&D inference compute (K) of total R&D spend (S_k) The inference spend share S_k , one of the two components of $\varepsilon_{R,C} = \gamma S_k$.	As aforementioned in the last subsection, we have almost no data on lab expenditure on inference or its share relative to other R&D inputs.	Ideal new data: Labs could publish detailed data on inference compute spend across domains and over time.
Tradeoff between training and inference compute (γ) The quality–quantity exponent γ , the other component of $\varepsilon_{R,C} = \gamma S_k$.	The parameter γ governs the returns to scaling capabilities C versus scaling inference compute K , since we model effective AI labor as $C^\gamma K$. Villalobos and Atkinson (2023) offers one estimate. They found that $\gamma \times \varepsilon_{C,A} = 1 \sim 2$, where $\varepsilon_{C,A}$'s estimation will be explained in subsection 3.3.	Current data weakness: the estimation is not up to date. Ideal new data: More extensive study on public benchmarks/with ECI.
The effect of AI use on researcher productivity A direct (if loose) measure of $\varepsilon_{R,C}$: how much capability raises effective R&D.	We have self-reported productivity gains from surveys: 2X from METR's developer survey (Becker 2026); 4X from the Mythos 5 system card. The magnitudes are hard to interpret, but the upward trend is clear. Favaro and Clark (2026) report lines of code merged per engineer per day rose roughly 8X between 2024 and Q2 2026.	Current data weakness: note that this is the productivity effect of AI, but ε_{RC} is the total research effort added due to one extra unit of capability. Ideal new data: an RCT that measures productivity benefits from extra capabilities.

Model primitive	Existing data	Data ask
The strength of autonomous R&D A direct measure of $\varepsilon_{R,C}$ (and the discovery-rate response $\varepsilon_{A,C}$): AI’s standalone contribution to discovery.	AI agents have recently advanced the frontier in some optimization problems; however, it is difficult to find a metric to measure their overall ability.⁵ Recent system cards report mixed results on benchmarks of AI R&D, generally outperforming humans on optimization problems when the humans are given 8 or 40 hours, but showing low pass rates on complex debugging problems (Mythos 5 system card, table 2.3.3.A; GPT-5.5 system card, section 9.1.3). These results are difficult to place on a common scale.	Current data weakness: benchmark questions are not the same as real-life LLM training problems. Ideal new data: How autonomous AI systems perform on real LLM training problems would be especially informative. The ideal data would characterize the returns to expenditure on agentic, human, and hybrid labor.
Capabilities of AI R&D A direct, qualitative read on $\varepsilon_{R,C}$ — which parts of research AI substitutes for, and where it remains weak.	Si et al. (2025) find that AI-generated AI research ideas are rated highly relative to human-generated ones. Trehan and Chopra (2026) finds a range of weaknesses, including both taste and execution. The Mythos system card has an extended discussion of the qualitative strengths and weaknesses of Mythos’ abilities at AI R&D.	Current data weakness: it remains unclear how to describe AI’s weaknesses in AI R&D relative to human researchers. A common split is between idea generation (taste) and idea execution. Ideal new data: Unstructured interviews with lab staff with well-designed questions eliciting AI R&D weaknesses.

3.3 Capabilities Response to Algorithmic Efficiency

The third term is the elasticity of capabilities with respect to algorithmic progress. Under the maintained assumption that capability depends on algorithmic efficiency A and training compute T only through effective compute, $C = C(A \times T)$, the two enter symmetrically, so $\varepsilon_{C,A} = \varepsilon_{C,T}$: the elasticity we care about equals the returns to capability from scaling training compute. This is precisely the object estimated by scaling-law studies, and Epoch’s work on the capabilities–compute relationship gives direct estimates (Ho et al., 2025; Epoch AI, 2025). We get estimates of around 6.5 when C is defined as e^{ECI} .

We treat this term as the best-measured of the three; the supporting model-training data is collected in the supplementary material (Section 3.4).

3.4 Supplementary Data

The data below does not measure the three terms directly but fills out the broader picture of AI R&D automation. The model-training rows support the third term (Section 3.3): they pin down the capability–compute scaling relationship. The same data also help identify historical growth in algorithmic efficiency and predict the effects of future growth in training compute and data. The final row covers the economic feedback loop, through which capability funds further inputs.

Model primitive	Existing data	Data ask
Model training laws Calibrates the capability map $C(AT)$; pins down $\varepsilon_{C,A} = \varepsilon_{C,T}$.	The basic model remains Chinchilla (Hoffmann et al., 2022), which maps data and compute to pretraining loss.	Current data weakness: Scaling laws for RL remain largely unpublished. Ideal new data: post-training compute usage of models.
Growth in training compute g_T ; since effective compute is AT , it drives capability growth.	Training compute has grown 4–5X per year since 2010 (Rahman and Owen, 2024). Whitfill et al. (2025) discusses possible GDP constraints on continued compute scaling.	Current data weakness: There is scarce data on model training compute since 2024. Ideal new data: training compute of new models.

Model primitive	Existing data	Data ask
Growth in data Data D, an input to capability beyond effective compute AT.	There are minimal existing estimates of data spend by labs.⁶	Ideal new data: labs could publish data spend, or partial estimates could be constructed from third-party data providers such as Surge AI, Scale AI, or other third-party data providers.
Share of AI’s economic value creation captured by AI labs Economic output Y, the channel through which capability funds further inputs (the economic feedback loop).	We can use firm revenue estimates, but these are likely to represent only a small share of the value created. Brynjolfsson et al. (2026) use online choice experiments to elicit willingness-to-accept for giving up generative AI chatbots, estimating that consumer surplus substantially exceeds firms’ revenues.	Current data weakness: More estimation needed, with time variance. Ideal new data: AI firms’ revenue overtime.

4 DISCUSSION

To assess the plausibility of self-sustaining acceleration we “calibrate” our analytical condition from Section 2 with the available estimates from 3. Certain parameters, namely the elasticity of algorithmic efficiency to AI capabilities, are unknown. Thus we derive a condition on the range of values for this elasticity that implies self-sustaining acceleration. We have a high degree of uncertainty in many of these empirical estimates, and also acknowledge the possibility of model misspecification. Thus, we also provide a general discussion of the most important points of evidence for and against self-sustaining acceleration.

4.1 Model calibration

Recall the self-sustaining acceleration condition from the bottlenecks model in Section 2.3:

$$\underbrace{\frac{\varepsilon_{\dot{A},R}}{1 - \varepsilon_{\dot{A},A}}}_{\text{return to R\&D}} \cdot \underbrace{\varepsilon_{R,C}}_{\text{research effort's elasticity to AI capabilities}} \cdot \underbrace{\varepsilon_{C,A}}_{\text{capabilities' elasticity to algorithmic progress}} > 1.$$

We plug in the following values:

- $\frac{\varepsilon_{\dot{A},R}}{1 - \varepsilon_{\dot{A},A}} = \frac{g_A}{g_R} \approx \frac{3}{3} = 1$. We take the $g_A = 3\times$ from Ho et al. (2024). We take $g_L = 3\times$ from Epoch AI (2026) and Whitfill and Wu (2025) and $g_E \approx 3\times$ from (You et al., 2026). This implies $g_R = 3$.
- $\varepsilon_{R,C}$. We have almost no evidence for this parameter given the improvement in capabilities in AI uplift studies is not well-reported.
- $\varepsilon_{C,A} \approx 6.5$. If we define capabilities as $\exp(ECI)$, we can get the elasticity from the slope of a regression of ECI on $\log(\text{compute})$ across models using data from Epoch AI.

Plugging in these values tells us that self-sustaining acceleration is achieved if $\varepsilon_{R,C} > .153$. This is the increase in R&D effort from an additional unit of ECI. For point of comparison, Claude Opus 4.8 is 1 ECI unit stronger than Claude Opus 4.7. Under our model, Anthropic would meet the self-sustaining acceleration condition if Claude Opus 4.8 lead to 15.3% higher research productivity than Opus 4.7. Note that the increase in research output per unit of AI capabilities tends to be smaller than the uplift for human tasks from AI use that is typically experimentally measured. This is because labor only accounts for a fraction of all research tasks.

Lastly, these elasticities may not be constant. In particular, we suspect research effort's elasticity to AI capabilities is currently increasing – the move from GPT-4 to GPT-5 led to a much larger increase in research productivity than the move GPT-1 to GPT-2. On the other hand, the marginal return to AI capabilities may eventually diminish. The continual measurement of those elasticities is necessary for understanding the current state of RSI.

4.2 Arguments for and against self-sustaining acceleration

Best evidence in favor of self-sustaining acceleration:

- The Mythos 5 system card showed a surprisingly high researcher uplift from AI use (self-reported 4x) and a 40-hour time horizon at which Mythos beat human researchers.
- Favaro and Clark (2026) assess the performance of human researchers versus Claude code in making AI research decisions. In April 2026, Claude Mythos Preview beat humans 64% of the time up from 50% of the time for Claude models released in 2025.

- The return to research effort for AI (1x as calculated in section 4.1) is higher than the return to research in the general economy (0.3 as estimated by Bloom et al., 2020).
- In August 2025, experts and superforecasters (METR, 2025)⁷ predicted an 8–20% chance that the growth rate of effective compute would triple by 2029. In May 2026, Jack Clark predicted a 60% chance that by the end of 2028, there will be “an AI system powerful enough that it could autonomously build its own successor” (Clark, 2026).

Best evidence against self-sustaining acceleration:

- Gundlach et al. (2025) argue that most algorithmic progress thus far has been scale-biased, i.e. returns have been increasing in compute, and have come from big breakthroughs such as the transformer. This suggests that scaling compute and “out-of-distribution” innovation will remain important.
- Compute bottlenecks could bind if experimental compute and cognitive labor are complements Whitfill and Wu (2025), and at the same time, productivity growth is not fast enough to support continued AI R&D investments, that is, if we “run out of GDP” (Halperin, forthcoming). Inelastic power markets could also limit continued scaling.
- The apple-picking model of AI R&D (Cunningham and Shetty, 2026): Terry Tao and Andrej Karpathy report that while models are useful at picking low-hanging research tasks, they quickly lose effectiveness on harder problems, suggesting diminishing returns—though this evidence remains early.

Important evidence that is ambiguous:

- Forecasts for acceleration are mixed (METR, 2025): the median AI expert believes there is a 21% probability of acceleration in algorithmic progress; none of the surveyed experts puts this probability above 50%.
- Millidge (2025) argues that “most algorithmic progress is data progress”, from better filtering of pre-training data to the substantial resources spent on post-training data. If true, this reduces the plausibility of explosive capabilities growth to the extent that data collection is difficult to automate. Pre-training filtering seems to be a one-time boost, whereas post-training data might be supplied more elastically and be amenable to automation. We are only beginning to learn how to build better reinforcement learning environments, and AI systems could help design future ones. And just as chain-of-thought (arguably an algorithmic innovation) was complementary to post-training data, future algorithmic improvements could be, too.

⁷“Experts and superforecasters both find a three-fold acceleration of effective compute scale-up by 2029 plausible but unlikely, with experts giving it a median 20% chance and superforecasters 8%.”

Narrow capabilities acceleration may come before broad capabilities acceleration. It seems plausible that AI R&D agents are effective at optimizing benchmarks but less effective at developing major breakthroughs that enable models to learn efficiently in new environments.⁸ Nonetheless, spillovers from narrow to broad progress remain possible. For example, an advanced narrow system, in pursuit of benchmark optimization, may discover new capabilities, such as highly sample-efficient algorithms able to update their weights during deployment, that quickly turn narrow progress into broader progress.

Political and social barriers. Even if explosive growth is technically plausible, political and social barriers could stop it in its tracks. Recent policy reactions to Mythos by the Trump administration, despite an earlier laissez-faire stance, suggest placing a greater weight on these barriers. If compute scaling remains important, even minor frictions in model deployment will slow revenue growth and further scaling.

4.3 Impacts of explosive capabilities growth on the world

Explosive capabilities growth differs from generic AI progress in *speed*: it could compress generations of progress into short periods. We highlight several aspects of explosive growth that distinguish it from gradual AI progress, and that may be especially important for understanding its social, economic, and political consequences.

- **Transition costs:** It seems plausible that transition costs increase with the speed of technological change. Labor markets are one such case: the costs of structural transformation may rise with its speed, as some economists argue was the case with the China Shock.
- **Institutional inertia:** In the world of explosive capabilities growth, institutions broadly defined — regulatory capacity, democratic procedures, legal frameworks, infrastructure, and so on — might lag behind (Bostrom, 2014; Ord, 2020). Rapid technological change might also grant political elites a temporary spike in *de facto* power, which they could use to capture institutions.
- **Asymmetric progress across domains:** As noted, we see self-sustained acceleration in narrow capabilities as more likely in the near term than an acceleration in broad capabilities. The former would likely be concentrated in tasks that are easy to verify in closed loops, such as software development, games, mathematics, and related areas. This could create acute asymmetries in progress across domains. For example, we may find solutions to long-standing puzzles in mathematics and breakthroughs in computational chemistry that enable new materials, alongside sluggish progress in less self-contained or messier domains.
- **Power imbalances across firms and nation-states:** Even model providers with relatively small gaps in technical skill could see vast gaps open up in effective capabilities and power over the world. Imagine Anthropic or the US government having

⁸Note this is becoming less true, as AI has contributed to groundbreaking math results.

access to nuclear-level capabilities that are vastly superior to those of any other actor.

- **Greater misalignment risk:** Misaligned AI might also create bigger risks under explosive capabilities growth. Sudden increases in model capabilities would provide less time for technical and social safeguards to be put in place.

4.4 Future Steps

Several directions for future work emerge, some of which we are actively pursuing:

- Derive the self-sustaining acceleration condition for the model with economic feedback loops.
- Measure the return to research off the balanced growth path.
- Distinguish quality from quantity improvements in data and compute.
- Extend the model to multiple research sectors (software, hardware, data).
- Model research taste and vision in AI R&D.
- Endogenize the decision to automate AI R&D.
- Incorporate directed technical change.

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APPENDIX

A SCALE-DEPENDENT ALGORITHMIC PROGRESS

Since algorithmic progress is usually not directly observable, it is typically estimated as the increase in capabilities not explained by increases in training compute, just as TFP is estimated by the Solow residual. That is, if we know (from a “scaling law”) that in the absence of algorithmic progress some index of AI capabilities C would track training compute T , it is conventional to propose the model

$$C = A \cdot T \tag{1}$$

and calculate A as C/T . Using a model like “our basic model” in Section 2.2, the resulting time series for A , in combination with a time series for research inputs R and the semi-endogenous AI R&D function, can then be used to generate a prediction for how fast algorithms would advance after the automation of AI R&D.

Gundlach et al. (2025) show conclusively that model (1) is misspecified, because algorithmic progress in AI development to date has been largely large-scale-biased. For an extreme case of large-scale-biased algorithmic progress, consider instead the model

$$C = \min(A, T) \cdot T. \tag{2}$$

Here, algorithmic improvements proportionally advance AI capabilities as long as there is the training compute to leverage them (i.e., here, as long as $T > A$). If we stopped scaling the compute, however, capabilities would ultimately stagnate (here, at T^2). Likewise, even if automating AI R&D yields a leap in the quality of the algorithms available, under (2) this will not deliver arbitrarily large advances in AI capabilities until we have accumulated the training compute to leverage these algorithms.

Another potential explanation for the Gundlach et al. finding, however, is that we have been developing large-scale-biased algorithms precisely *because* we have been scaling compute. Indeed, we might imagine living in a somewhat different world in which we knew exactly what architectures will take us to superintelligence, but are simply too compute constrained to make use of such architectures efficiently. Why don’t we live in that world? An answer is simply that we have no incentives to discover such algorithms—and would not be able to validate them if we did—in a world in which we are compute constrained. This also suggests that if we did face a long-term compute constraint, or if an AI model were tasked with recursively self-improving on fixed compute, we might develop algorithms that made ever better use of “small” training runs—just as academic labs to some extent do today. That is, rapid RSI may still be feasible, but only via some kind of ‘directed technical change’.